Zach Fisher

zachfisher.com ✦ linkedin.com/in/zkf ✦ me@zachfisher.com

Staff Software Engineer at Google DeepMind working on LLM post-training, including reward modeling, synthetic data, and scientific-agent evaluation. Ph.D. physicist with a track record of bringing scientific rigor to frontier-model research and production ML.

Employment

Google DeepMind (formerly Google Research) — *Mountain View, CA*
Staff Software Engineer — *May 2025 – Present*
Senior Software Engineer — *April 2022 – April 2025*
Software Engineer — *April 2020 – April 2022*

- Built reward models for reinforcement learning post-training of frontier LLMs, including rubric-conditioned critiques, code-based auto-raters, and critic-driven self-improvement methods. Contributions were incorporated into multiple Gemini training revisions.
- Conceived and led development of a scientific-agent benchmark for long-horizon optimization with real-world simulators. Designed dozens of tasks across wide-ranging engineering and science domains, built the evaluation infrastructure and leaderboard from scratch, evaluated frontier models, and identified an exploration gap between leading model families. The benchmark was adopted for Gemini internal leaderboards.
- Designed and scaled a failure-mode-driven synthetic data generation pipeline for frontier model post-training across SQL, data science, mathematical reasoning, and scientific coding. The system generated targeted training curricula with automated multi-stage quality filtering and simulator-backed verification.
- Diagnosed and resolved critical failures in frontier training runs, including silent infrastructure failures, reward signal degradation, evaluation flakiness, and distribution shifts in model behavior. Built real-time health dashboards and sample-exploration tools to accelerate live-run triage.
- Led research on simulator-driven reinforcement learning for agentic tool use, designing a procedural data generation framework that produced dense-reward inverse problems across 5 scientific domains for RL post-training.
- Led work on attribute extraction models for Google Shopping and Maps, processing tens of millions of edits across 80+ attributes at over 96% precision and increasing recall by 3–5× across four languages.
- Enabled partner teams across Search, Maps, and post-training by sharing models, evaluation frameworks, and infrastructure. Mentored engineers and researchers across organizations and taught an internal machine learning course.

Perimeter Institute for Theoretical Physics — *Waterloo, ON, Canada*
Postdoctoral Researcher — *September 2017 – February 2020*

- Conducted research in quantum field theory, quantum information, and quantum gravity as an It from Qubit postdoctoral fellow. Published work on black hole information, holography, entropy bounds, and quantum energy conditions.

Education

University of California, Berkeley — *August 2012 – May 2017*
Ph.D. in Physics. Thesis: "Entropy Bounds and Entanglement." Co-formulated the Quantum Null Energy Condition and Quantum Focusing Conjecture, and co-authored the QNEC's first proof, connecting quantum information inequalities to spacetime energy conditions.

Massachusetts Institute of Technology — *September 2008 – June 2012*
S.B. in Physics. Thesis: "Shuttling of ions for characterization of a microfabricated ion trap."

Selected Publications

- H. Wu, **Z. Fisher** et al. "Automated Optimization of Structured Large Language Model Prompts via Iterative Evaluation." Technical Disclosure Commons (2026).
- Y. Sheng, S. Gandhe, B. Kanagal, N. Edmonds, **Z. Fisher**, S. Tata, A. Selvan. "Measuring an LLM's Proficiency at Using APIs: A Query Generation Strategy." KDD-ADS (2024).
- B. Gunel, J.B. Wendt, J. Xie, Y. Zhou, N. Vo, **Z. Fisher**, S. Tata. "STRUM-LLM: Attributed and Structured Contrastive Summarization." arXiv:2403.19710.
- R. Aksitov, S. Miryoosefi, Z. Li, D. Li, S. Babayan, K. Kopparapu, **Z. Fisher**, R. Guo, S. Prakash, P. Srinivasan, M. Zaheer, S. Kumar. "ReST meets ReAct: Self-Improvement for Multi-Step Reasoning LLM Agent." arXiv:2312.10003.
- S. Ontanon, J. Ainslie, V. Cvicek, **Z. Fisher**. "Making Transformers Solve Compositional Tasks." ACL (2022).
- J. Ainslie, S. Ontanon, C. Alberti, V. Cvicek, **Z. Fisher**, P. Pham, A. Ravula, S. Sanghai, Q. Wang, L. Yang. "ETC: Encoding Long and Structured Inputs in Transformers." EMNLP (2020).
- R. Bousso, **Z. Fisher**, S. Leichenauer, A. Wall. "Quantum Focusing Conjecture." Phys. Rev. D 93, 064044 (2016).
- R. Bousso, **Z. Fisher**, J. Koeller, S. Leichenauer, A. Wall. "Proof of the Quantum Null Energy Condition." Phys. Rev. D 93, 024017 (2016).

Awards

- Product Achievement Award and Publication Excellence Award, Google Research / Google DeepMind.
- It from Qubit Postdoctoral Fellowship, Simons Collaboration on Quantum Fields, Gravity and Information.
- Editors' Suggestion, Physical Review D, for work on quantum entropy bounds.

Skills

- **Core engineering:** Python, JAX, PyTorch, TensorFlow, Apache Beam, SQL, C++, distributed training, production ML debugging.
- **ML systems:** RL post-training, RLHF, reward modeling, value modeling, model evaluation, agent evaluation, tool use, synthetic data generation, long-context agents.
- **Research domains:** scientific AI, simulator-backed evaluation, quantum information, quantum gravity, entropy bounds, mathematical reasoning.